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Neogene paleomagnetism and oroclinal bending of the central Andes of Bolivia

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Abstract. Geophysical and tectonic evidence suggests that the principal vertical axis bending of the Bolivian Orocline is a late Cenozoic event related to subduction and Andean uplift. To further test this hypothesis, paleomagnetic data are presented from Miocene basins within the Altiplano and Cordillera Oriental of Bolivia, located in the presumed southern limb of the orocline. To factor out potential VGP data, samples were collected from sedimentary sections spanning at least 0.5 m.y. New paleomagnetic data are presented for two sites, i.e., Cerdas (21°S latitude, 16-15 Ma) and Quehua (20°S latitude, 13-7 Ma). In conjunction with previously published paleomagnetic results from middle to late Tertiary rocks from Bolivia and Peru spanning some 16° of latitude (6-22°S), a consistent pattern of oroclinal development is as follows: significant counterclockwise rotation (about 15-20°) in the northern limb, negligible rotation in the axis (at the latitude of the Arica Deflection), and significant clockwise rotation (about 15-20°) in the southern limb. Although these paleomagnetic data could represent independent, mesoscale (>10^2 to <10^4 km^2) block rotations, the preferred interpretation is that they confirm the hypothesis that a significant portion (about 30-40°) of the oroclinal bending occurred during the late Cenozoic. Using the unit mean paleomagnetic declinations in the well-dated sections, minimum rates of oroclinal bending during the late Cenozoic are calculated to range from 0.6°/m.y. to 1.7°/m.y. The remaining 15-20° of observed curvature of the orocline probably resulted from an earlier phase of Andean deformation or may represent the shape of the ancestral Andes.

Introduction

In plan view many of the orogenic zones of the world are distinctly arcuate, and classically, it has been assumed that this reflects the original continent-wide shape of the mountain range. In contrast, *Carey* [1955] proposed the concept of an orocline to describe large-scale, vertical axis bending of mountain ranges that have occurred during orogenesis. *Carey* [1958] also coined the term "Bolivian Orocline" to describe the approximately 45° bend observed in the central Andes at the so-called "Arica Deflection," which is the presumed oroclinal axis. Since Carey's initial ideas, many papers have been written either supporting or challenging the orocline hypothesis, as will be discussed further below. In the context of plate tectonics, proponents of the Bolivian Orocline [e.g., *Allenby*, 1987; *Isacks*, 1988] have stated that a major phase in this oroclinal bending occurred during the late Cenozoic resulting from Andean orogeny and subduction of the Nazca plate. While many tectonic and structural models and data have been used in discussions of oroclines, paleomagnetic studies offer some of the most tangible evidence for the magnitude and direction of rotations in tectonically active regions [*Eldredge et al.*, 1985; *Beck*, 1989].

The extensive sedimentary and volcaniclastic deposits within high-elevation basins of the Altiplano and Cordillera Oriental of Bolivia are centrally located within the central axis and southern limb of the presumed orocline, but until recently they have been an untapped resource for paleomagnetic data. Most of these basins are of middle and late Cenozoic age [*Hoffstetter*, 1986; *Marshall and Sempere*, 1991] and contain thick, relatively continuous outcrops with numerous fossil faunas and volcanic ashes. As such, these deposits potentially provide well-calibrated tests of tectonic rotations within the Andean orogen.

The purpose of this paper is twofold. (1) New paleomagnetic and geochronological data are presented for two Miocene sedimentary sequences, Cerdas and Quehua (Figure 1), which are located at 21° and 20°S latitude, respectively, in the southern limb of the Bolivian Orocline. (2) These results are then combined with previously published data for Cenozoic rocks from Bolivia (Figure 1) and Peru and analyzed as they relate to the

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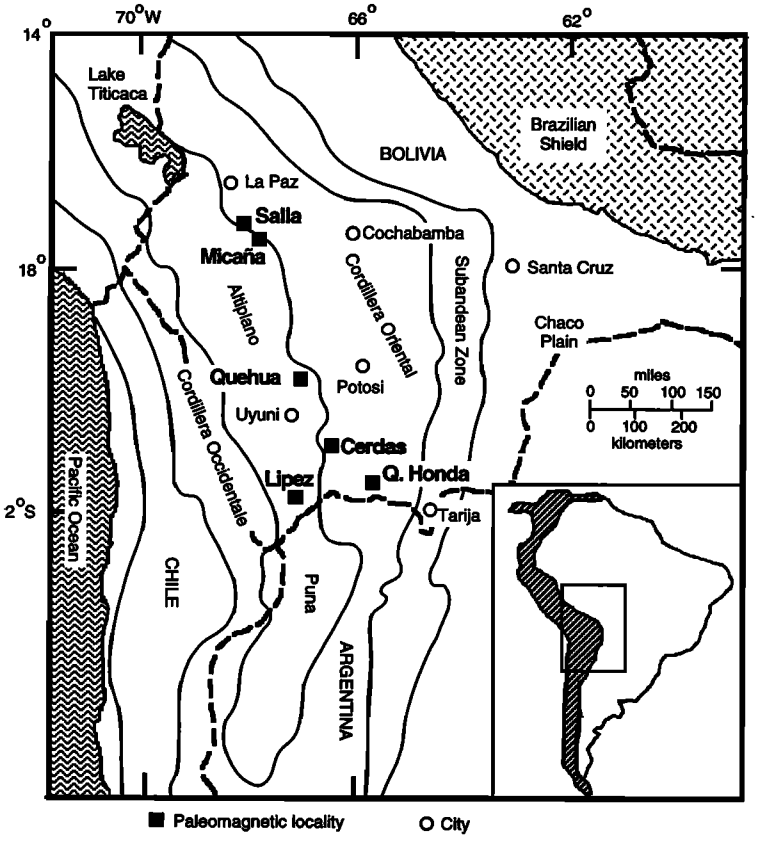


Figure 1. Location map showing the Neogene localities discussed in this report from the southern limb and axis of the Bolivian Orocline. New data are presented here for Quehua and Cerdas; data are also presented for previous studies from Salla, Micaña, and Quebrada Honda [*MacFadden et al.*, 1990] and Lipez [*Roperch et al.*, 1993].

hypothesis of the development of the orocline during the Neogene. The prediction is that paleomagnetic vectors, or so-called "declination anomalies," from late Tertiary sediments lying in the southern limb of the orocline will show a pattern of uniform, clockwise, vertical axis rotations, whereas those from the northern limb will show counterclockwise rotations, both resulting from deformation during the later phase of Andean orogeny.

Bolivian Orocline and Andean Tectonics

There is a diversity of opinion about the origin of the arcuate shape of the Andes and whether or not it fits the notion of oroclinal bending first proposed by *Carey* [1955, 1958] and later refined in light of plate tectonic theory [e.g., *Isacks*, 1988]. There are essentially two schools of thought that exist with regard to the origin of the shape of the Andes, i.e., oroclinal bending versus mesoscale in situ block rotation. A related question is the timing of the Andean orogeny, as is also discussed below.

Oroclinal Bending

As mentioned above, *Carey* [1955, 1958] proposed the concept of an orocline for a mountain range that is bent during orogeny and used the Andes as a prime example. Thus in the initial stages of the formation of the Andes the major trace of the mountains was relatively straight and aligned roughly N-S; bending ensued during orogenesis. Carey did not explicitly state a mechanism for so-called "oroclinal rotation," other than that it resulted from "compressive forces," but he believed that most of the bending occurred in the northern limb north of the Arica Deflection; the latter is the presumed oroclinal axis. Since Carey's presentation of the Bolivian Orocline hypothesis, data from numerous geological and geophysical studies have been used to address the timing and development of this major tectonic feature. Because they can measure both the magnitude and direction of vertical axis rotations, paleomagnetic studies in the Andes have been frequently cited in discussions of the Bolivian Orocline, as well as elsewhere. *Isacks* [1988] summarized the then available paleomagnetic results, most of which were from Mesozoic rocks. These data generally show a consistent pattern of about 15-20° counterclockwise and clockwise rotations, respectively, north and south of the oroclinal axis (although see other interpretations below), thus accounting for much of the 45° bend at the Arica Deflection.

It is generally accepted that the major phases (including Incaic and Quechua) of Andean orogeny have occurred during the late Cenozoic [e.g., *James*, 1971; *Jordan et al.*, 1983; *Isacks*, 1988; *Sempere et al.*, 1990]. Accordingly, while the relatively large body of Mesozoic paleomagnetic data estimates the magnitude of rotation, it only generally constrains the timing of this process; younger data are needed to resolve this issue. Paleomagnetic data from the Neogene Orcos dyke swarms from Peru [*Heki et al.*, 1983,

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1984, 1985; *Kono et al.*, 1985] were among the first results used to more precisely constrain the timing of counterclockwise oroclinal bending in the northern limb (although see *Beck*, 1988 and *Mitouard et al.*, 1990). One could argue that being from a single volcanic sequence, these results may represent a VGP (virtual geomagnetic pole) or non-time-averaged reading of the ancient geomagnetic field. The data from the Ocros dyke swarms have since been corroborated by other studies from the presumed nothern limb of the orocline in Peru including (1) about 25° of counterclockwise rotation from early Oligocene rocks [*Mitouard et al.*, 1990], (2) about 16.5° of counterclockwise rotation from post-Eocene rocks [*Macedo-Sánchez et al.*, 1992a], and (3) about 19° of counterclockwise rotation from middle Eocene to middle Miocene volcanic rocks [*Macedo-Sánchez et al.*, 1992b] (note that some of these rotational data are recalculated, as described below). In addition, *Roperch et al.* [1993] report about 14° of counterclockwise rotation from Oligo-Miocene sediments from Puerto Japones and Chama in Bolivia; at 16-17°S latitude these localities are in the transitional zone between the northern limb and axis of the orocline.

Paleomagnetic data for the presumed southern limb of the Bolivian Orocline include (1) about 18° of clockwise rotation from middle Miocene sediments at Quebrada Honda [*MacFadden et al.*, 1990] and (2) about 39° of clockwise rotation from the 20 Ma Rondal volcanics at Lipez [*Roperch et al.*, 1993]. Lying within the presumed oroclinal axis, paleomagnetic data from Salla [*MacFadden et al.*, 1985, 1990] indicate negligible rotation during the past 25 m.y. Another paleomagnetic data set from the late Miocene (about 6-7 Ma) Micaña locality [*MacFadden et al.*, 1990] in the same region indicates either (1) negligible rotation in the oroclinal axis or (2) deposition after significant vertical axis rotation had occurred.

Thus most of the available paleomagnetic data indicate about 15-20° of Neogene rotation uniformly distributed within the presumed limbs of the orocline with negligible rotation in the axis. If the absolute values of Neogene rotation are summed from both limbs, then the total of between 30° and 40° can account for most of the bend in the Arica Deflection and is consistent with the hypothesis that the development is related to late Cenozoic orogeny [*Allenby*, 1987; *Isacks*, 1988]. Estimates of the amount of flexure at the Arica Deflection range from 45° [e.g., *Isacks*, 1988] to 60° [*Beck*, 1987], depending upon how and where one places a protractor on the map. With either estimate, anywhere from 5° to 20° of oroclinal bending occurred before the mid-Tertiary; alternatively, this may well represent the ancient shape of the primordial Andes.

It should be noted that not all paleomagnetic data are consistent with a model of, or interpreted as supporting, oroclinal rotation in the Andes. For example, *Scanlan and Turner* [1992] report counterclockwise rotations of about 10° from Jurassic and Cretaceous units in northern Chile at 19°S latitude. Scanlon and Turner interpret the magnitude and lack of significant rotations implied by the data from these units, which lie in the presumed southern limb of the orocline, to be inconsistent with the model of the Neogene Bolivian Orocline. Perhaps these represent an earlier rotation not part of the Neogene phase of oroclinal development, or, depending upon their structural context, they may represent local, in situ, mesoscale block rotation unrelated to the continent-wide tectonic development of the Andes. Likewise, *Beck et al.* [1986] and *Hartley et al.* [1992] also report paleomagnetic data from Cretaceous rocks that are interpreted to represent block, and not oroclinal, rotations. In a comprehensive study of fault motions, *Marrett et al.* [1994] found no structural evidence for rotation in the foreland region of the southern Puna (latitude 22-27°S) of Argentina, and they argue that the 10-30° rotation seen to the north cannot account for the approximately 40° necessary to account for observed crustal shortening in the southern Andes.

In Situ, Block Rotation

As suggested in the preceding paragraph, there is by no means a consensus on the interpretation of relevant data and acceptance of the concept of Neogene oroclinal bending of the Andes. Consistent with other tectonic models, another hypothesis suggests that the Arica Deflection and arcuate trace of the Andes are ancient features and that continent-wide oroclinal bending never occurred, or has been negligible. *Scanlan and Turner* [1992] suggest that this original arcuate shape resulted from a primary pattern of ridge-ridge-ridge triple junction during the Mesozoic separation of Gondwanaland. In this model, the two roughly NW and S trending ancient ridges form the original trace of the Andes, and the third ridge formed an alaucogen [*Burke and Dewey*, 1973], or a so-called "failed arm" of a triple junction, trending roughly eastward from the axial region of the Andes. *Fletcher and Litherhand* [1981] suggested that the Mesozoic Velasco Alkaline Province in eastern Bolivia, which trends at high angle (to the northeast) from the Andes, represents an alaucogen remaining from the initial rifting of Gondwanaland.

The triple junction model, however, does not explain the roughly equal and opposite counterclockwise and clockwise Neogene rotations seen in, respectively, the northern and southern limbs of the presumed orocline. Opponents of the oroclinal bending theory explain these paleomagnetic data to represent local, in situ, block rotations resulting from distributed shear [*McKenzie and Jackson*, 1989] throughout the Andes and not resulting from a single, continent-wide mechanism of uniform oroclinal bending [*Beck*, 1989]. In a more complex model, *Dewey and Lamb* [1992] suggest that the Andes can be separated into four separate regions, each with different sets of structural and tectonic processes that have occurred, and taken together, they do not support a model of uniform oroclinal bending throughout this mountain range.

If oroclinal bending did not occur, then what is the scale of in situ, block rotations? There are structural models and data from other tectonic settings which suggest that the size of these blocks are on the order of 10^4 km^2 [*Beck*, 1989; *Jackson and McKenzie*, 1989; *Luyendyk*, 1989]. Unfortunately, at present the paleomagnetic data that could be used to evaluate the scale and magnitude of presumed in situ block rotations are not yet available from Cenozoic rocks in the Andes; however, with the availability of further data sets in the future, these tests might be possible.

The above discussion presumes that the models of vertical axis oroclinal bending versus mesoscale block

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rotation are mutually exclusive. To a large extent, this is the way that they have been presented for the Andes. However, in a recent paper, *Beck et al.* [1994] combine both models to explain the observed paleomagnetic rotations in the Mesozoic and Cenozoic rocks of the Andes and relate these to a mechanism driven by plate convergence and oblique subduction.

Paleomagnetic Methods and Results

The initial reason for collecting paleomagnetic samples from several Andean basins of Bolivia was because of their long, relatively continuous and undeformed sedimentary sequences that could be used to calibrate the highly fossiliferous record of extinct mammals [e.g., *MacFadden et al.*, 1985, 1990]. It was soon realized that the paleomagnetic data derived from these sequences were relevant to tectonic studies because of the following attributes. (1) A precise chronology of the samples in question is available using a combination of radioisotopic, biostratigraphic, and magnetostratigraphic data. (2) With long, relatively continuous sections and flat-lying or gently dipping sediments, the small-scale (i.e., <10 km^2) structural context is less complex than in highly deformed terranes, e.g., as is characteristic of pre-Cenozoic rocks in the Andes. (3) Each of the sections span at least 0.5 m.y., and, as such, they remove any potential error introduced in VGP data, i.e., those that are a short-term reading of the ancient geomagnetic field (as is the case with single, rapidly cooling volcanic units). The field and laboratory procedures were similar for the two new localities discussed below, i.e., Cerdas and Quehua. Accordingly, the paleomagnetic field and laboratory methods and results for both localities are discussed here. The interpretation of the magnetostratigraphy and correlation to the magnetic polarity time scale (MPTS) are presented separately below after the discussion of the $^{40}Ar/^{39}Ar$ age determinations.

As is standard magnetostratigraphic procedure [*Lindsay et al.*, 1987], three separately oriented paleomagnetic samples were collected from each site, or horizon, within the stratigraphic sections studied. A total of 34 and 32 separate paleomagnetic sites were collected from, respectively, Cerdas and Quehua. The average vertical separation between sites is about 6 m, and sufficient fine-grained lithologies were found so that there are no gaps of more than 10 m in the sections sampled. Each section was chosen because it contained known fossil mammals or potentially dateable volcanic ash horizons (also see below). In the laboratory the paleomagnetic samples were cut into standard 2.5-cm cubes and analyzed in the Paleomagnetics Laboratory at the University of Florida. Magnetizations were measured in a 2G cryogenic magnetometer, and samples were treated using Schonstedt static alternating field (AF) and thermal demagnetizers. All measurements were made in a magnetically shielded room.

Several previous studies of the paleomagnetic characteristics of Cenozoic fluvio-lacustrine sediments of the Bolivian Andes indicate that both fine-grained magnetite and hematite are the dominant carriers of the remanence, although additional contributions in the low unblocking temperature components can originate from goethite and sometimes magnetic sulphides [*MacFadden et al.*, 1990, 1993; *Quidelleur et al.*, 1992]. Given this potential magnetic mineralogy, demagnetization using a minimum of 10 (thermal) to 13 (AF) steps was done for each sample (some were demagnetized up to 22 steps). In general, two samples from each site were thermally demagnetized in steps from 200 to 600°C (and higher, if warranted), and the third was subjected to AF demagnetization in steps of 5 to 70 mT. If the direction of magnetization was different in the AF-demagnetized sample, it was then subjected to a minimum of seven stepwise thermal demagnetizations from 300 to 600°C. The presumed magnetic mineralogy and magnetic vector components of the sediments analyzed are exemplified from the demagnetization characteristics presented in Figure 2. In many cases (e.g., Figures 2c and 2e) thermal demagnetization revealed a low unblocking temperature component that was removed at temperatures at, or below, 200°C; this appears to be goethite. AF demagnetization removed a low-coercivity component in fields of about 10 mT (e.g., Figure 2b); this is probably a viscous overprint within magnetite. A midrange unblocking temperature component is found in most samples between about 300° and 575°C; this probably results from a combination of magnetite and hematite. A midrange coercivity component between about 20 and 60 mT is apparent in many of the samples treated with AF demagnetization; this probably results from magnetite. Sometimes AF demagnetization was ineffective in isolating a stable component (e.g., Figure 2b, vertical component); further treatment with thermal demagnetization frequently resulted in linear decay to the origin implying a stable vector component. At temperatures above 550-575°C, i.e., in the range to be expected from hematite, magnetic directions either were similar to those observed for the midrange (e.g., Figure 2e) or became erratic; in other samples (e.g., Figure 2c), there was a significant drop in intensity indicating that most of the stable component of magnetization was carried by magnetite. These observations are also borne out by the relative drop in the mean magnetic intensities for all samples; natural remanent magnetization (NRM) = 7.39 x 10^{-4} A/m (N=197); 30 mT = 0.56 x 10^{-4} A/m (N=37); 60 mT = 0.35 x 10^{-4} A/m (N=37); 300°C = 1.99 x 10^{-4} A/m (N=164); 450°C = 0.77 x 10^{-4} A/m (N=193); 550°C = 0.48 x 10^{-4} A/m (N=176); and 600°C = 0.29 x 10^{-4} A/m (N=92).

In the majority of samples, linear decay segments were selected from vector demagnetization plots in the midrange of thermal demagnetization (i.e., 300-575°C) and analyzed for the least squares, best fit vector component [*Kirschvink*, 1980]. For other samples in which linear decay segments did not contain sufficient points, did not trend toward the origin, or calculated maximum angular deviation (MAD) values were greater than 20, then a single step, usually between 300 and 550°C, was chosen to represent the characteristic direction of magnetization. Each magnetic vector thus chosen was analyzed using Fisher statistics [*Fisher*, 1953] to obtain a site mean direction. Of the 67 originally collected sites from the Cerdas and Quehua, 49 were class 1 and five were class 2 or 3 [after *Opdyke et al.*, 1977]; the remaining sites exhibited poor magnetic characteristics and were not used for subsequent data interpretation. As will be discussed for each of the two localities presented below, the class 1, 2,

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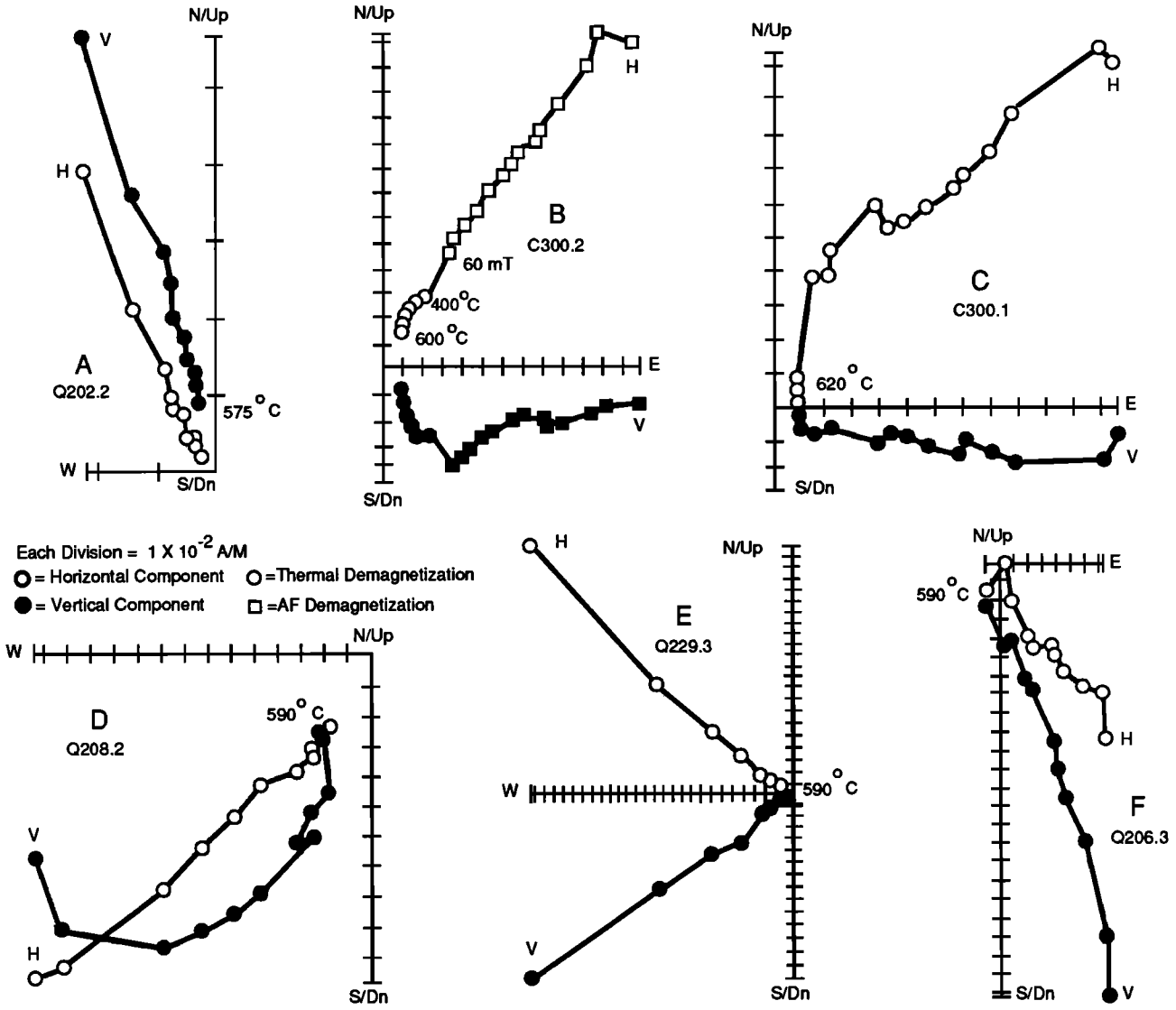


Figure 2. Orthogonal vector demagnetization ("Zijderveld") diagrams showing the behavior of magnetic vector components (horizontal and vertical) using thermal (Figures 2a and 2c-2f) and alternating field demagnetization (Figure 2b).

and 3 data were used to interpret the magnetostratigraphy (54 sites), whereas only class 1 data (49 sites) were used to calculate formational or section mean data and paleomagnetic rotations using Fisher statistics.

The relative scatter of the NRM measurements versus the demagnetized samples (Figure 3) and a positive reversals test, with overlapping mean vectors from the normal and reversed populations (Figure 4), indicate that the laboratory regime effectively isolated characteristic components of the ancient geomagnetic field at, or near, time of deposition. (Another standard stability criterion, the fold test, was not done for either locality because the sediments are either flat lying or of such shallow dip that this test was not possible.) In summary, it is concluded that this regimen of stepwise laboratory treatment effectively isolated a stable characteristic component of the NRM that could be used to interpret site polarity and directions of formational mean rotations as will be presented below in the separate sections dealing with Cerdas and Quehua.

$^{40}Ar/^{39}Ar$ Geochronology

Fourteen ash samples were collected in the field at the same time that paleomagnetic sites were sampled and the sections were measured and described. As such, their stratigraphic positions relative to the other descriptions and data from Cerdas and Quehua are unambiguous. As available, the tuffaceous samples were collected throughout each section to provide maximum and minimum age determinations that could then be used to calibrate the fossil mammal horizons and magnetostratigraphy.

The minerals separated from the ashes were loaded into wells of an Al sample disk along with a centrally located monitor mineral and were wrapped in Al foil and a thin layer of Cd. The package was then irradiated for 30 min with a fast neutron fluence of 5.7 x 10^{13} neutrons/cm^2 in the hydraulic rabbit facility core of the Omega West research reactor at Los Alamos National Laboratory.

Following irradiation, individual grains were loaded into

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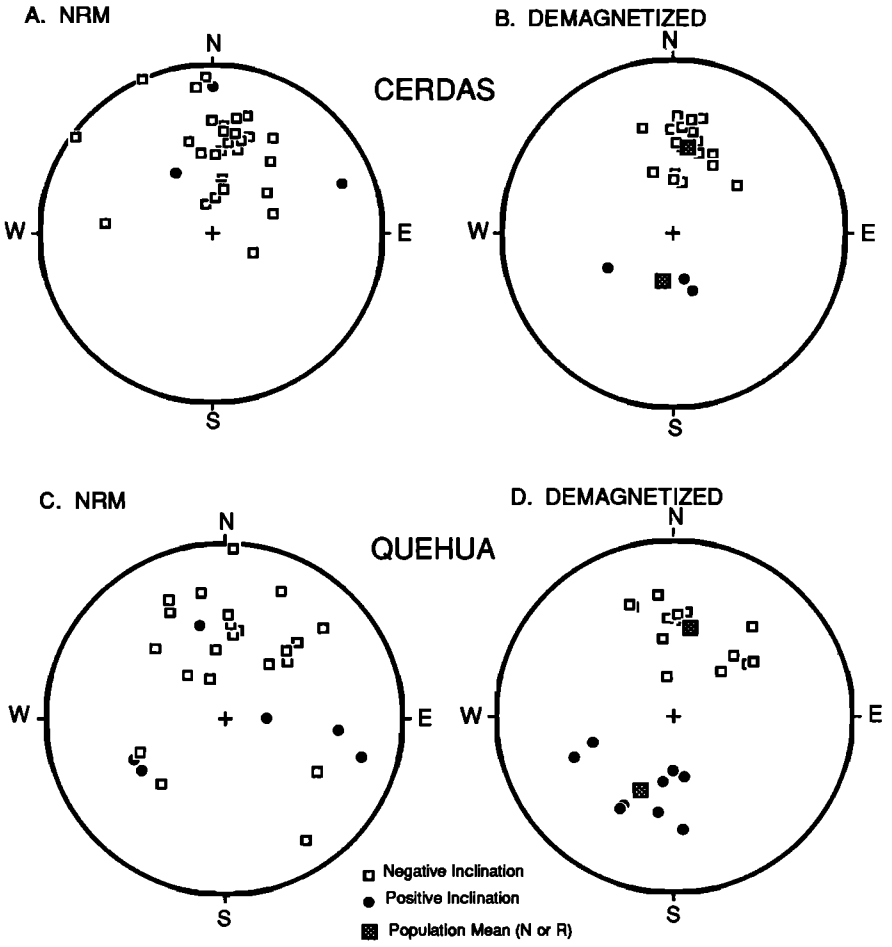


Figure 3. Site means for all class 1 sites for Cerdas and Quehua showing (a) and (c) natural remanent magnetization directions and (b) and (d) directions after demagnetization.

wells of a copper sample disk, enclosed within the sample chamber, bolted onto the extraction system, and baked out at 200°C for 8 hours. Total fusion of minerals was accomplished with a 6W coherent Ar ion laser. The released gases were then purified by two Zr-Fe-V getters operated at approximately 150°C. Argon was measured in an on-line Mass Analyzer Product 215 noble gas mass spectrometer operated in static mode and using automated data collection techniques. Sample fusion, gas purification, and mass spectrometry were automated with computer-programmed schedules.

Ages were calculated using a J value of $3.57 \pm 0.003 \times 10^{-4}$ as calculated from replicate analyses of individual grains of the co-irradiated monitor mineral Fish Canyon Tuff sanidine with a reference age of 27.84 Ma (modified from *Cebula et al.* [1986]) intercalibrated in-house with Minnesota hornblende MMhb-I at 520.4 Ma [*Samson and Alexander*, 1987]. Ca and K corrections used during this study were determined from laboratory salts: $(36/37)Ca = 2.53 \times 10^{-4} \pm 5 \times 10^{-6}$, $(39/37)Ca = 6.38 \times 10^{-4} \pm 1.0 \times 10^{-5}$, and $(40/39)K = 2.4 \times 10^{-3} \pm 7.0 \times 10^{-4}$. Mass discrimination as determined from replicate analyses from an on-line pipette system during this study yields a mean value of 1.008 ± 0.0004. Decay constants are those recommended by *Steiger and Jaeger* [1977] and *Dalrymple* [1979].

An abridged summary of the results of $^{40}Ar/^{39}Ar$ analyses is presented in Table 1. (All analytical data are presented in the electronic data supplement.[1]) Of the 14 samples originally collected from Cerdas and Quehua, 13 age determinations from nine of the ashes (using a combination of biotite, plagioclase, or sanidine) are interpreted to be reliable age estimates for the associated stratigraphic sequence. It also should be noted that in most of these cases the ages are internally consistent, i.e., they are progressively younger up each stratigraphic section.

[1]An electronic supplement of this material may be obtained on a diskette or Anonymous FTP from KOSMOS.AGU.ORG. (LOGIN to AGU's FTP account using ANONYMOUS as the user name and GUEST as the password. Go to the right directory by typing CD APPEND. Type LS to see what files are available. Type GET and the name of the file to get it. Finally, type EXIT to leave the system.) (Paper 95JB00149, Neogene paleomagnetism and oroclinal bending of the central Andes of Bolivia, by B. J. MacFadden, F. Anaya, and C. C. Swisher III). Diskette may be ordered from American Geophysical Union, 2000 Florida Avenue, N. W., Washington, D. C. 20009; $15.00. Payment must accompany order.

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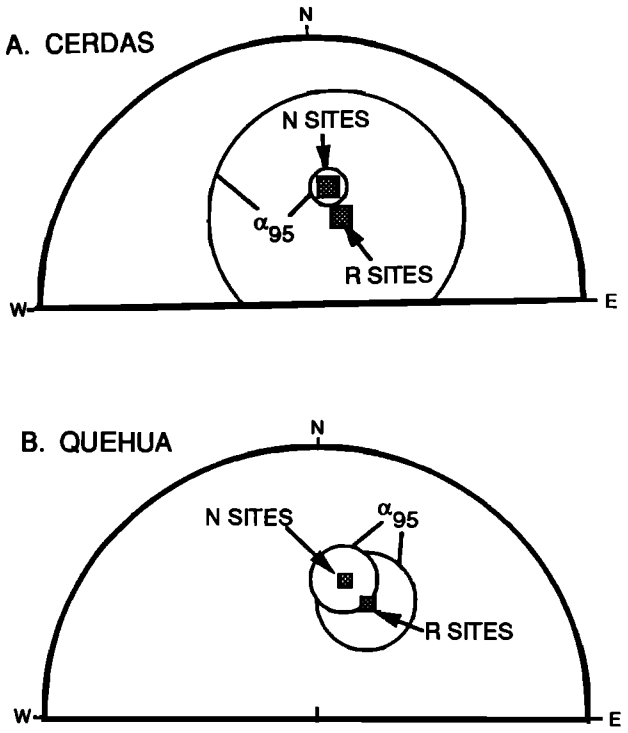


Figure 4. Stereographic projections of normal and reversed site population means and associated α_{95} circles of confidence for (a) Cerdas and (b) Quehua. The reason for the large α_{95} for the reversed sites from Cerdas is because the population consists of only three site means.

New Miocene Paleomagnetic and Geochronologic Results

Until recently, the Andean basins of Bolivia have been relatively poorly studied in contrast to, for example, the classic Cenozoic exposures of Argentina. Starting in the 1960s, fossil mammals were recovered from several sites in the Altiplano and Cordillera of Bolivia [*Hoffstetter*, 1968; *Hoffstetter et al.*, 1972; *Villarroel*, 1974]. At about the same time, volcanic samples were collected that provided initial calibrations for some of these basin sequences [e.g., *Evernden et al.*, 1977]. With the advent of magnetic polarity stratigraphy and the presence of long, relatively continuous and highly fossiliferous sections, the chronology of middle and late Cenozoic Andean basins of Bolivia advanced significantly. *MacFadden et al.* [1985] and *Naeser et al.* [1987] presented the chronology of Salla, one of the older, i.e., late Oligocene/early Miocene basin deposits. There are several important middle Miocene localities, including Quebrada Honda, Micaña, Cerdas, and Quehua [*Villarroel*, 1978; *Hoffstetter*, 1977; *MacFadden et al.*, 1990], which are calibrated by a combination of biostratigraphic, paleomagnetic, and/or radiometric results. Later Miocene and Pliocene localities on the Altiplano and Cordillera Oriental of Bolivia are likewise abundant and now relatively well calibrated [e.g., *Hoffstetter*, 1986; *Marshall et al.*, 1992; *Marshall and Sempere*, 1991]. Thus within the past few decades, knowledge of the precise ages and sequence of high-elevation sedimentary basins of Bolivia has improved to a point at which this sequence can be used to better understand the geological history of this tectonically active region.

Table 1. $^{40}Ar/^{39}Ar$ Age Determinations From Cerdas and Quehua, Bolivia

Sample	Mineral (Number of analyses)	Stratigraphic Location	Age ± SE, Ma
		Cerdas	
91CS-CA3	biotite (6)	about 50 m below base of section 1	16.305 ± 0.045
91CS-CA3	biotite (6)	about 50 m below base of section 1	16.358 ± 0.071
91CS-CA1	biotite (6)	145 m above base of section 2	15.105 ± 0.073
		Quehua	
90CS-Q202L	biotite (6)	13 m above base of section 2	11.764 ± 0.040
90CS-Q202L	sanidine (2) and plagioclase (3)	13 m above base of section 2	12.611 ± 0.034
90CS-Q202U	biotite (7)	13 m above base of section 2	12.277 ± 0.040
90CS-Q202U	sanidine (3)	13 m above base of section 2	12.704 ± 0.055
90CS-Q2	biotite (7)	37 m above base of section 2	12.681 ± 0.168
90CS-QU01	biotite (5)	below base of section 1	9.243 ± 0.059
90CS-QU01	plagioclase (7)	below base of section 1	11.099 ± 0.111
90CS-QU1	sanidine (2)	8 m above base of section 1	8.911 ± 0.038
90CS-QU3	sanidine and plagioclase (8)	"Toba Chiu Chiu," 55 m above base of section 1	8.518 ± 0.024
90CS-QU5	biotite (7)	3 m below top of section 1	6.844 ± 0.386

An expanded version of this table is provided as an electronic data supplement.

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Cerdas

Cerdas (also Cerdas-Atocha) consists of a large series of badlands located at 21°52'S latitude, 66°18-19'W longitude (Gran Chocaya 1:50,000 quadrangle map), Departmento de Potosi, in the southern altiplano, about 60 km southeast of the city of Uyuni (Figure 1). The middle Tertiary sediments in this region have been mapped as the Quehua Formation [e.g., *Marshall and Sempere*, 1991], although the exact relationships to typical rocks of this name located some 125 km to the north are somewhat unclear. Fossil mammals have been reported from the Cerdas area [*Villarroel*, 1978; also *Hoffstetter*, 1986], and on the basis of the stage of evolution of the notoungulates, primarily *Plesiotypotherium minus*, the sediments in this area were thought to be middle Miocene age, or roughly about 12 Ma. *Marshall and Sempere* [1991] state that this faunule is thus Chasicoan land mammal age (about 11.5-9 Ma), although because of endemism in Bolivia relative to typical faunas from this age in Argentina, characteristic taxa are lacking. The Cerdas badlands are moderately to highly fossiliferous, and it is not possible to place the exact location of the fossils described earlier into our stratigraphic sections (see below).

We measured three sections at Cerdas that appear to be relatively continuous and contain fossil mammals and interbedded tuffaceous horizons (Figure 5). These localities approximate the general locality from which *Villarroel* [1978] described the presence of the notoungulates *Plesiotypotherium minus* and *Microtypotherium* cf. *M. choquecotense*. The composite section has a thickness of about 200 m, is flat lying or dipping at less than 2°, and consists primarily of brown and reddish-brown silty clays, silts, and silty sands of predominantly fluvio-lacustrine origin. Numerous interbedded, lenticular gravels also occur at Cerdas, particularly in the upper part of Section 3; these were obviously not sampled for paleomagnetism. Known fossil mammal horizons occur throughout the section, with a particularly rich zone (yielding about 50 dentitions, mostly *Plesiotypotherium minus*, in 1 day of collecting) located at 140 m above the base of the composite section.

We collected tuffaceous samples for potential

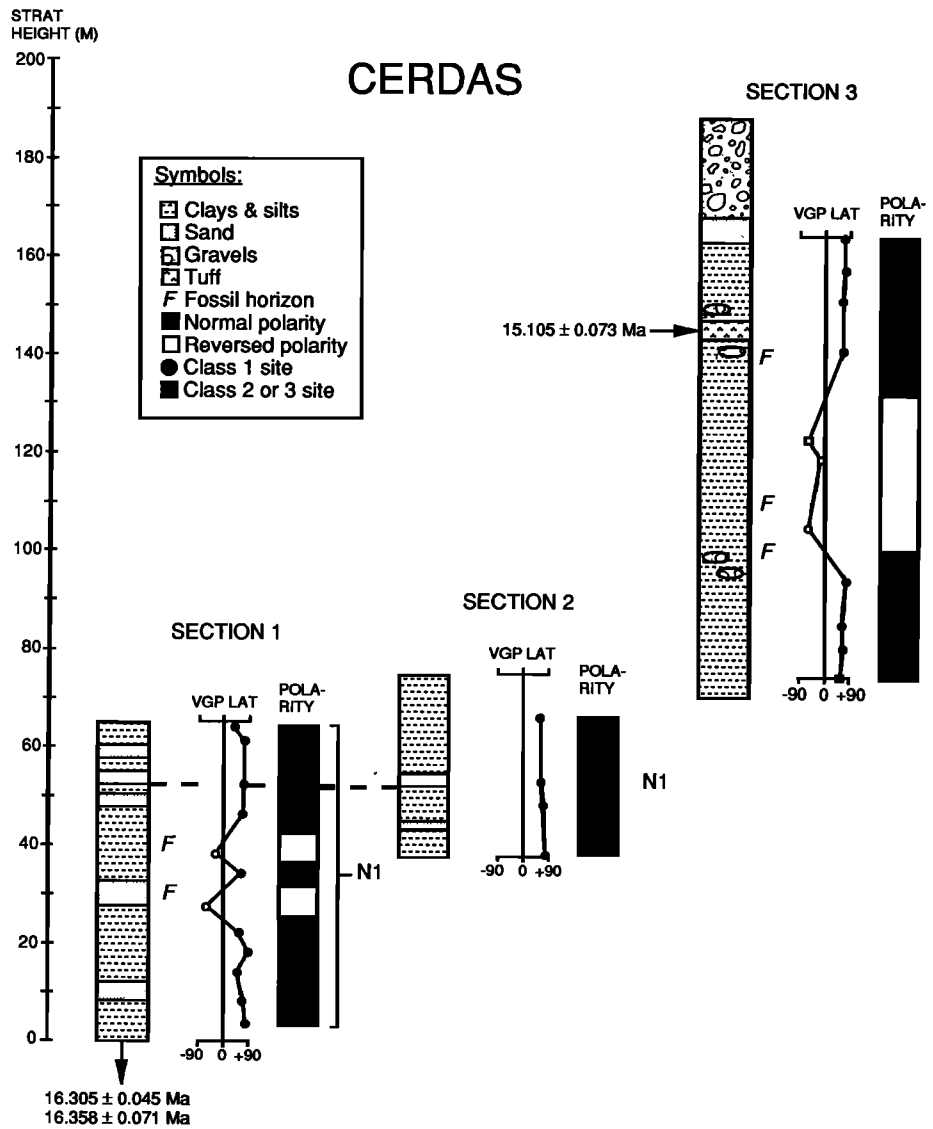


Figure 5. Lithostratigraphy and magnetostratigraphy, fossil horizons, and $^{40}Ar/^{39}Ar$ age control for the three measured sections at Cerdas.

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radioisotopic age determinations from four horizons associated with the three measured sections. Subsequent laboratory analyses yielded $^{40}Ar/^{39}Ar$ age determinations of 16.305 ± 0.045 Ma and 16.358 ± 0.071 Ma from about 50 m below the base of section 1 and 15.105 ± 0.073 Ma from the 145-m level of section 3 (Figure 5 and Table 1). Thirty-four paleomagnetic sites were collected from the measured sections. After laboratory and statistical analyses, the 24 class 1 sites yield a paleomagnetic mean direction of declination 10.1°, inclination -39.1° (α_{95} = 7.0°, k = 18.9°; reversed sites flipped to northern hemisphere, also see Table 2; all class 1 individual site means for Cerdas and Quehua are available as an electronic data supplement available from AGU). Twenty-seven sites (24 class 1, three class 2 or 3 [after *Opdyke et al.*, 1977]) are used to interpret the magnetic polarity stratigraphy. Using these results, three principal magnetic zones are represented, i.e., N1, R1, and N2. Two class 1 single-site reversed polarity sites located in the lower third of the section initially were not considered reliable in correlating to the MPTS (although see below). Given the fragmentary nature of the fossil mammal fauna as yet known, and the recognized problems of correlating middle Miocene mammals in South America using the existing land mammal chronology [e.g., *Flynn and Swisher*, 1995], only the $^{40}Ar/^{39}Ar$ data are used to constrain the correlation of the magnetic polarity zonation to the MPTS. With these constraints and using the time scale of *Cande and Kent* [1992], zone N1 from the base to the 100-m level correlates to chron 5Cn of the MPTS (Figure 6), zone R1 to the intervening chron 5Br, and the upper part of the sampled section above the 140-m level to 5Bn.2n. The

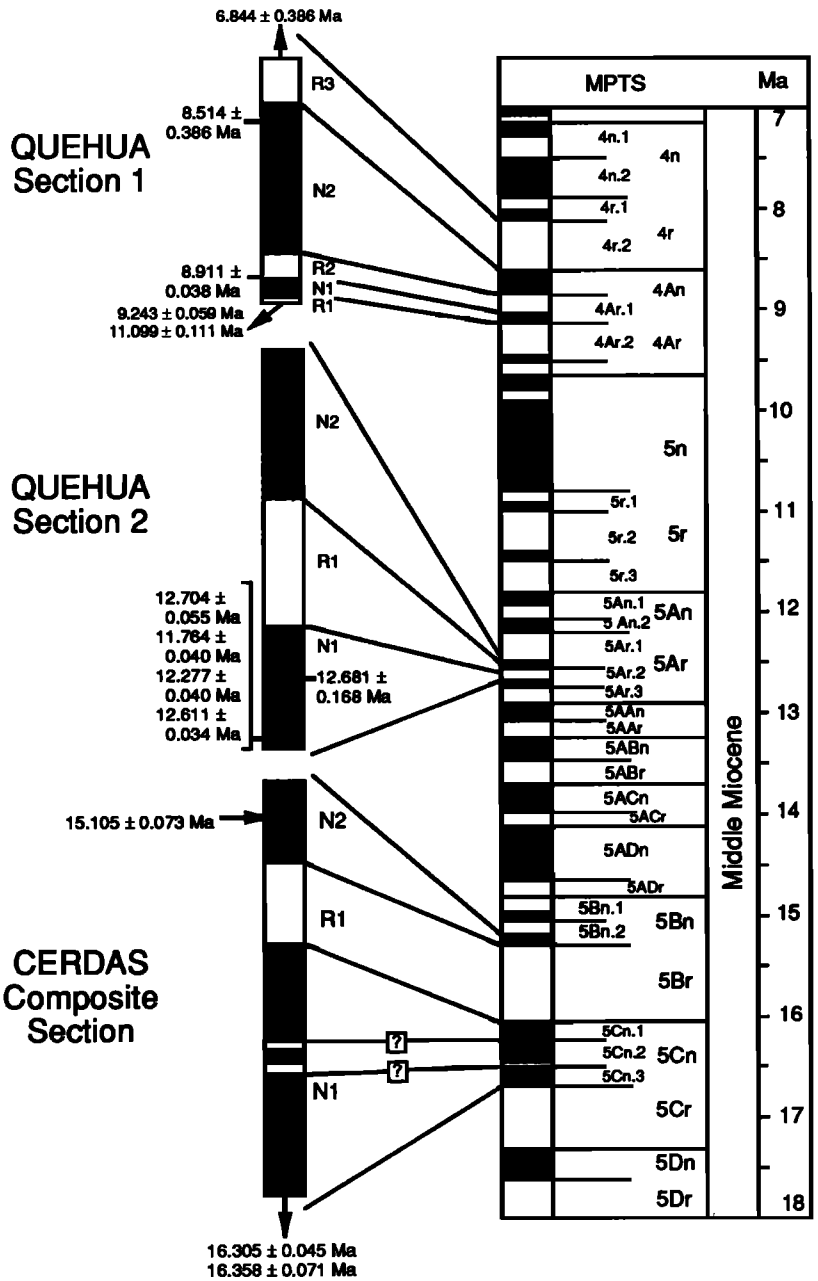


Figure 6. Correlation to the magnetic polarity time scale (MPTS) [*Cande and Kent*, 1992] for Cerdas and Quehua.

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two class 1 single-site reversed subzones within N1 may represent short-duration (about 100,000 years or less) events within chron 5Cn of the MPTS. Except for the two possible reversed events in zone N1, this correlation is unambiguous, i.e., given the associated error limits on the $^{40}Ar/^{39}Ar$ age determinations, it is the only one that fits with the radioisotopic data. As such, the composite section from Cerdas spans an age from about 16.5 to 15.3 m.y.

Quehua

Quehua consists of badland exposures located at 19°56'S latitude, 66°53'W longitude (Quehua 1:50,000 quadrangle map), Departmento de Potosi, in the southern altiplano, about 80 km north of the city of Uyuni (Figure 1). These badlands, and most other middle Tertiary sedimentary sequences on the southern altiplano, have been mapped as Quehua Formation, of which a lower and upper part is generally recognized. However, the lack of lithological continuity and precise calibrations across such a large geographic region have obscured the utility of this stratigraphic unit [e.g., *Villarroel*, 1978]. Fossil mammals have been reported from Quehua, and on the basis of biochronology, *Hoffstetter* [1986] correlated the typical Quehua Formation to the late Miocene, between about 10 and 6 m.y. On the basis of the stage of evolution of the notoungulate *Plesiotypotherium*, *Villarroel* [1978] considered the lower Quehua Formation to be about 25 Ma and the upper Quehua Formation to be about 20 Ma, although he realized that these correlations were only approximate and could be several million years in error. *Richter et al.* [1992; Appendix C] consider the lower Quehua to be early Miocene, calibrated by a $^{40}K/^{40}Ar$ age of 18.2 ± 0.2 Ma; and it spans an age from about 20 to 15 m.y., whereas the upper Quehua is middle Miocene and spans an age from about 15 to 10 m.y.

We measured two stratigraphic sections in the area of Quehua, i.e., (1) the 100-m-long section 1 containing the distinctive Chiu Chiu tuff ("toba") and (2) the 240-m long section 2 containing several superposed tuffaceous and fossil mammal horizons (Figure 7). Both sections consist predominantly of brown and reddish brown silty clays, silts, and silty sands of predominantly fluvio-lacustrine origin; section 1 also contains several freshwater limestone units. The uppermost part of section 2 grades into poorly indurated sands, which contain fossils (e.g., a large toxodont) but were not amenable to paleomagnetic sampling. The structure of the sections is straightforward with gentle dips of about 15° in the bottom of section 2 to subhorizontal elsewhere. Fossil mammals are found at several stratigraphic horizons within our measured sections, and although they are abundant, specimens are not found as frequently as they are in the exceedingly rich horizon at Cerdas.

We collected tuffaceous samples for potential radioisotopic age determinations from eight horizons within both measured sections. Of these, six produced reliable analytical results (Table 1). Section 1 is well constrained by ages of 11.099 ± 0.111 Ma and 9.243 ± 0.059 Ma below the base, 8.911 ± 0.038 Ma at the 8-m level, 8.518 ± 0.024 Ma at the 55-m level (corresponding to the "Toba Chiu Chiu"), and 6.844 ± 0.386 Ma at the 95-m level near the top. Two horizons within the lower 40 m of section 2 yielded five age determinations ranging from 12.704 ± 0.055 Ma to 11.764 ± 0.040 Ma (Table 1). Unfortunately, no other tuffaceous horizons were available to constrain the age of the upper part of section 2. Thirty-three paleomagnetic sites were collected from sections 1 and 2. After laboratory and statistical analyses the 25 class 1 sites yield a paleomagnetic mean direction of declination 15.3°, inclination -37.9° (α_{95} = 9.3°, k = 10.7; reversed sites flipped to northern hemisphere; also see Table 2). Twenty-seven sites (25 class 1 and two class 2 or 3) are used to interpret the magnetostratigraphy. Section 2 consists of three magnetic polarity zones, N1, R2, and N2. As with Cerdas, the fragmentary fossils are not helpful in a precise correlation of these magnetic zone to the MPTS; thus the radioisotopic ages are used for this purpose. Given the $^{40}Ar/^{39}Ar$ ages, magnetic zones N1 through N2 most probably correlate to the normal portions of chron 5Ar on the MPTS (5Ar.2 and 5Ar.1, respectively) with the intermediate R1 correlating to 5Ar.2 [*Cande and Kent*, 1992] (Figure 7). Section 2 thus spans an age range between 12.8 and 12.6 m.y. It should be noted that this is the most probable correlation, but the busy nature of the MPTS during this time and the lack of a radiosiotopic calibration points higher in the section leaves open the possibility of other correlations for magnetic zones N1-N2, e.g., the about 0.5 Ma younger chron 5An).

Section 1 consists of five magnetic polarity zones, R1-R3, which are based on a total of nine paleomagnetic sites. Using the two $^{40}Ar/^{39}Ar$ age determinations from within the part of section 1 containing the magnetic polarity zonation, there is a unique correlation to the late part of chron 4Ar (local zones R1 through R2), chron 4An (local zone N2), and the earliest part of chron 4r (local zone R3). Using this correlation thus yields a duration of between about 9.3 to 8.4 m.y. for the lower and middle (i.e., up to 75 m height) parts of the section. Above 75 m there is no magnetic polarity zonation, and there is a distinct lithological change in sedimentation to a regime of less clastics and more lower energy freshwater limestones. The $^{40}Ar/^{39}Ar$ age determinations of 6.844 ± 0.386 Ma at the 95-m level is consistent with the magnetic polarity zonation below and correlation to the MPTS. However, this significantly younger date suggests the presence of greatly reduced rates of sedimentation in the top 20 m and/or the possibility of periods of erosion and/or nondeposition.

Neogene Paleomagnetic Data and Development of the Bolivian Orocline

In addition to their importance for magnetostratigraphy, the paleomagnetic data from Cerdas and Quehua are important because they contribute to a understanding of the rotational history of the central Andes during the late Cenozoic. The new paleomagnetic results from Quehua and Cerdas as well as those from relevant previous studies are presented in Table 2. These data span some 16° of latitude and include the southern and northern limbs and the axis of the Bolivian Orocline. We assert that, when considered together, these data confirm the model proposed by *Isacks* [1988] in which a principal phase of oroclinal rotation occurred during the Neogene. The prediction from the model is that there should be roughly equal amounts of rotation on both limbs of the orocline with little rotation in the axial region, i.e., the Arica Deflection.

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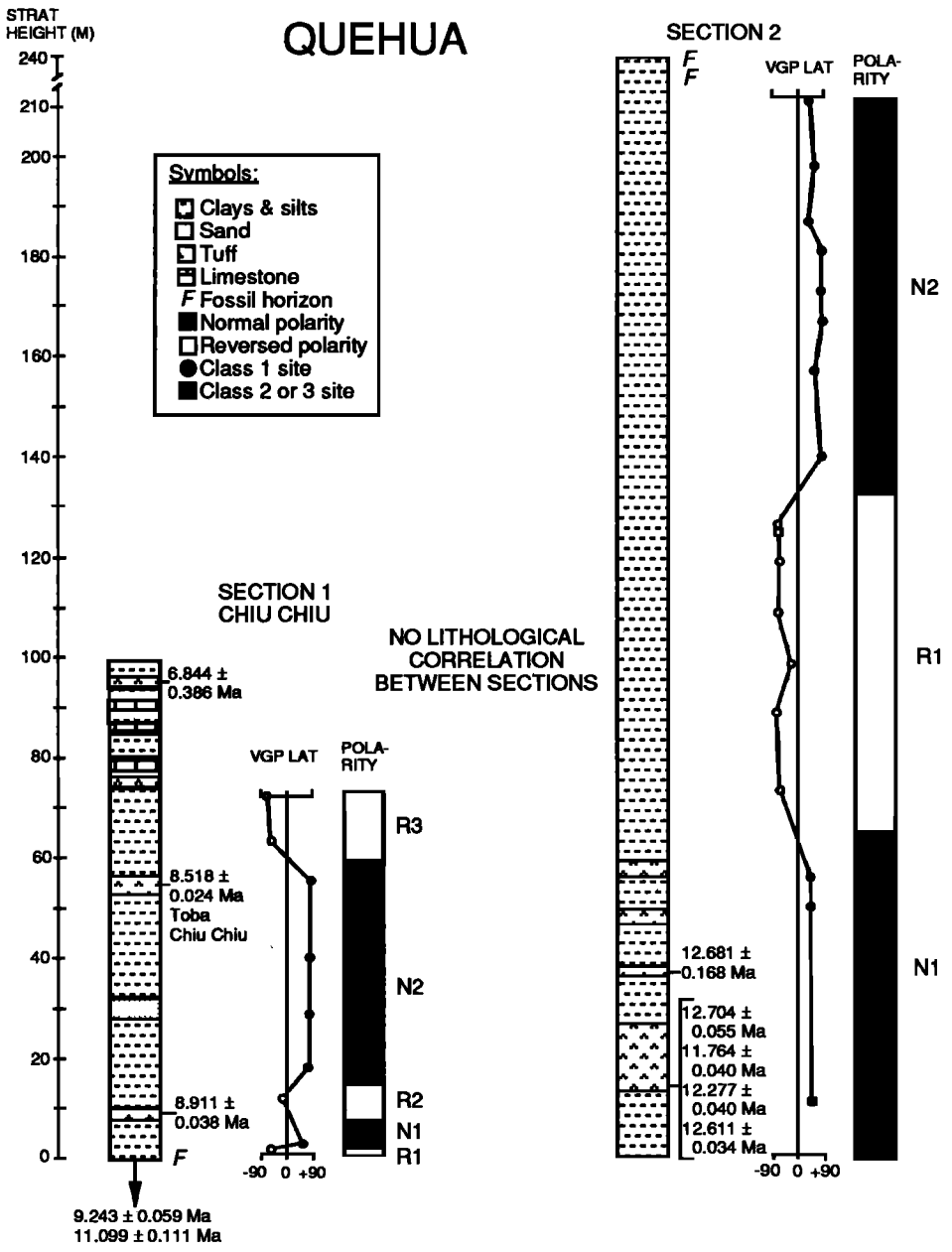


Figure 7. Lithostratigraphy and magnetostratigraphy, fossil horizons, and $^{40}Ar/^{39}Ar$ age control for the two measured sections at Quehua.

The calculation of "declination anomalies" ($D \pm \Delta D$) and the amount of rotation ($R \pm \Delta R$) for the South American paleomagnetic data has been the topic of considerable discussion in the literature. While everyone would agree that the unit or formational mean declinations are the primary data of relevance, these must be compared to a South American reference pole in order to assess net declination and rotation about a vertical axis. The apparent polar wander path for South America is not well constrained for the critical time interval with regard to the development of the Bolivia Orocline. There have been several solutions to this problem. (1) *Heki et al.* [1985] and *Kono et al.* [1985] conclude that for the late Neogene, the present-day pole should be used. (2) Although they noted that Cenozoic reference poles are poorly defined, *Macedo-Sánchez et al.* [1992a] nevertheless used the 15 Ma pole of *Valencio and Creer* [1968] and 5 Ma pole of *Valencio and Mendia* [1974] to calculate the amount of rotation for volcanic rocks in Peru. (3) For middle Tertiary rocks, *Mitouard et al.* [1990] interpolate between the 50 Ma and late Tertiary poles presented by *Irving and Irving* [1982]. (4) In a method similar to 3, *Beck et al.* [1994] use a 50 Ma "anchor" point and interpolate between this pole and the present-day dipole to calculate a reference pole for the different-aged rocks in question. (5) *Roperch et al.* [1993] present amounts of rotation that are different from their unit mean declinations, implying comparison with a South American reference pole; however, the latter is not specified.

Given the fact the relevant data for the southern limb of the orocline are of Neogene age and that there is no concensus about which method should be used, we follow

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Table 2. Neogene Paleomagnetic Data Relevant to the Development of the Bolivian Orocline

Locality	Latitude, °S	Longitude, °W	Age Ma	Number[a] of Sites	Mean[b] D, deg	Mean[b] I, deg	α_{95}, deg	Rotation[c] ($R \pm \Delta R$), deg	Reference
				Southern Limb					
Quebrada Honda	22	65	13-12	79	17.8	-40.7	3.9	17.8 ± 4.1	*MacFadden et al.* [1990]
Lipez	22	67	20	15	39	-38	11	39 ± 11.0	*Roperch et al.* [1993]
Cerdas	21	66	16.5-15.5	24	10.1	-39.1	7.0	10.1 ± 7.2	this study
Quehua	20	67	13-7	25	15.3	-37.9	9.3	15.3 ± 9.5	this study
				Oroclinal Axis: Arica Deflection					
Micaña	17.5	67	7.5-6.5	25	355.2	-25.8	5.9	-4.8 ± 5.2	*MacFadden et al.* [1990]
Salla	17	67	28-24	58	353.4	-37.4	5.4	-6.6 ± 5.4	*MacFadden et al.* [1990]
				Northern Limb					
Pto Japones -Chama	16-17	68.5	Oligo-Miocene	12	346	-34	10	-14 ± 9.7	*Roperch et al.* [1993]
Ocros	13	74	8.5	32	345.8	-31.8	5.3	-14.2 ± 5.0	*Heki et al.* [1983, 1985]
Various volcanics	11-13	76-78	post-Eocene	9	343.5	-29	8.6	-16.5 ± 7.9	*Macedo-Sánchez et al.* [1992a]
Various volcanics	11-15	76-77	Eocene-Miocene	14	341.4	-27.3	4.7	-18.6 ± 4.2	*Macedo-Sánchez et al.* [1992b]
Cajamarca	6-7	79	early Oligo-cene	20	333.4	-20.6	5.7	-26.6 ± 4.9	*Mitouard et al.* [1990]

[a]Number of sites used to calculate unit or formational mean direction.

[b]All vectors are flipped to northern hemisphere; D, mean, I, inclination.

[c]Positive and negative indicate, respectively, clockwise and counterclockwise rotations. Using the present-day reference pole (see discussion in text), the amount of rotation R = Mean D, and the error uncertainty is calculated using $\Delta R = 0.8\,\Delta D$, where $\Delta D = \sin^{-1}(\sin(\alpha_{95})/\cos(I))$ [*Demarest*, 1983; *Butler*, 1992]. R and ΔR may differ from previous studies depending upon reference pole originally used.

Heki et al. [1983, 1984, 1985], *Kono et al.* [1985], and *MacFadden et al.* [1990] and use the present-day reference pole; thus the observed declination anomaly is the amount of rotation [*Eldredge et al.*, 1985]. In addition, the data from previous studies have been recalculated (Table 2) using the present-day reference pole in order to allow uniform comparisons of the amount of rotation for each of the individual localities. When future studies agree upon a definitive late Tertiary reference pole for South America, at the very least the results presented in Table 2 will stand as a comparison of the relative amount of rotation for different parts of the Bolivian Orocline and can be subsequently recalculated from the original data.

Given the discussion above concerning the uncertainties and inconsistent use of a late Cenozoic paleomagnetic reference pole for South America, we would like to emphasize the following point. We assume that the South American reference pole coincides with the present-day pole and therefore interpret the declination anomalies to indicate the amount of vertical axis tectonic rotation within the Bolivian Orocline.

The new paleomagnetic data from Cerdas and Quehua further confirm those presented for Miocene sediments from Quebrada Honda [*MacFadden et al.*, 1990] and for the Miocene Rondal volcanics from Lipez, Bolivia [*Roperch et al.*, 1993] in which the southern limb of the orocline has undergone clockwise rotation within the past 15 m.y. (Figure 8). The values of rotation ($R \pm \Delta R$ [*Demarest*, 1983; *Butler*, 1992]) range from 39° ± 11.0° to 10.1° ± 7.2° with a simple (i.e., nonspherical) mean of 20.6° for the four localities, although the extreme value from the volcanic unit may be an outlier (due to the possibility of a VGP). Taking into account only the three sedimentary units, all of which span 0.5 m.y. or greater, the simple mean rotation is 14.4°. Paleomagnetic data from the axial region (Arica Deflection) are represented by two sites, Salla and Micaña [*MacFadden et al.*, 1990]. With an age of about 28-24 Ma and taking into account the error uncertainty (ΔR), Salla indicates that there has been negligible rotation in the axial region of the Bolivian Orocline. Given the younger age (7.5-6.5 Ma), the paleomagnetic data from Micaña indicate either negligible rotation or that the significant portion of the rotation occurred before the latest Miocene. In conjunction with the data from Salla, the first possibility for interpretation of the Micaña is preferred, although several recent studies have concluded that the principal phase of vertical axis rotation related to Andean orogeny ended by about 10 Ma [e.g., *Tsunakawa et al.*, 1987; *Gubbels et al.*, 1993].

The northern limb of the orocline is represented by five

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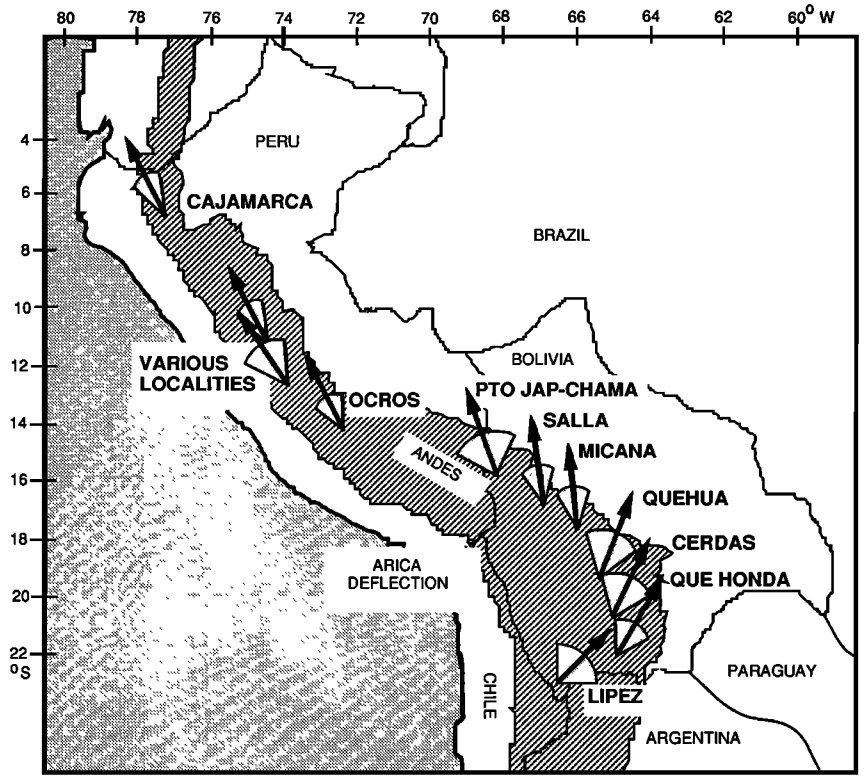


Figure 8. Map with plots of mean amounts of paleomagnetic rotation for each locality in the northern and southern limbs and axis of the Bolivian Orocline. Cones around the mean vector, R, represent the error uncertainty, ΔR (also see Table 2).

Cenozoic paleomagnetic data sets. The original data presented by *Heki et al.* [1983, 1985] ΔR is recalculated here) indicate -14.4° ± 5.0° (although *Beck* [1988] reinterprets the paleomagnetic data and associated errors to indicate negligible rotation) of vertical axis, counterclockwise rotation (note that a negative sign before the rotation indicates counterclockwise). Four other studies of Cenozoic rocks of Peru and northernmost Bolivia have subsequently been presented that are also relevant to an understanding of the northen limb of the orocline. (1) *Mitouard et al.* [1990] present data from early Oligocene rocks between 6 and 7°S latitude indicating a rotation of -26.6° ± 4.9°. (2) *Macedo-Sánchez et al.* [1992a] present paleomagnetic data from nine separate volcanic flows of post-Eocene age spanning some 2° of latitude with a rotation of -16.5° ± 7.9°. (3) *Macedo-Sánchez et al.* [1992b] report paleomagnetic data from Eocene to Miocene volcanic rocks that indicate a rotation of -18.6° ± 4.2°. (4) *Roperch et al.* [1993] report -14° ± 9.7° of rotation from Oligocene and Miocene rocks of northernmost Bolivia. The mean declination anomalies from these studies range from -26.6° ± 4.9° to -14° ± 9.7° with a simple mean of -18.0°. These paleomagnetic results are therefore indistinguishable, although in the opposite sense, from the amount of rotation observed from the southern limb of the orocline in Bolivia.

In summary, the paleomagnetic data synthesized here confirm the prediction of roughly equal amounts of oroclinal rotation during the Neogene [*Isacks*, 1988]. Although the majority of the data do not have sufficient temporal precision (e.g., "post-Eocene"), those from the four localities in the southern limb (Cerdas, Quebrada Honda, Quehua, and Lipez) and one from the northern limb (Ocros, Table 2) are sufficiently well constrained with radioisotopic ages to allow estimates of the minimum rate of oroclinal bending. Using the mean age and R for each locality, the estimates of rotation rate range from 0.6° /m.y. for Cerdas, to 1.7° /m.y. for Ocros. These rates are minimum estimates assuming that bending has continued up to the present day. If oroclinal rotation ended by the late Miocene, as has been suggested [e.g., *Sempere et al.*, 1990; *Gubbels et al.*, 1993], then the actual rates were much greater.

If rotational data from the north and south limbs of the Bolivian Orocline are summed, then this amount accounts for a significant portion, i.e., about 30-40° of the bend seen at the Arica Deflection. Other data from Cretaceous rocks summarized elsewhere [e.g., *Heki et al.*, 1984; *Beck*, 1987, 1988; *Beck et al.*, 1994] likewise suggest rotation possibly resulting from an earlier phase of oroclinal bending. Our data support the concept of Neogene oroclinal bending related to later phases of Andean uplift and subduction of the Nacza plate; however, the lingering question remains about the possibility that these data instead indicate mesoscale block rotation resulting from distributed shear within the Andes [e.g., *Beck*, 1988]. While this possibility cannot be ruled out entirely, the identification of the scale of the block rotations has remained enigmatic for the Andes. If it is on the order of 10^4 km^2, as *Luyendyk* [1989] has seen in southern California, then a model needs to be proposed to account for counterclockwise, negligible, and clockwise rotation

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from north to south in the Andes. This is perhaps the case given the study by *Beck et al.* [1994] that unifies both models and asserts that oblique subduction can provide the tectonic mechanism to explain the process of Andean orogenesis. Additional studies of paleomagnetic rotations in the Andes can further test the hypothesis of oroclinal bending as well as evaluate alternate interpretations for this interesting continent-wide orogenic zone.

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